

RYBINSK, Russian Federation

WMO: 272250

Lat: 58.100N

Lon: 38.683E

Elev: 105

StdP: 100.07

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.7	-23.4	-29.1	0.3	-26.6	-25.7	0.4	-23.3	9.0	-6.4	8.0	-5.4	1.5	310	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	28.6	20.8	26.7	19.8	24.9	18.8	21.7	26.9	20.6	25.4	19.5	23.8	2.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.7	24.7	18.7	13.7	23.3	17.7	12.9	22.4	63.6	26.9	59.8	25.6	56.1	24.0	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.4	DB	-30.1	31.0	3.2	1.9	-32.4	32.3	-34.3	33.4	-36.0	34.5	-38.4	35.8	(4)
(5)			WB	-30.2	22.9	3.1	1.7	-32.5	24.1	-34.3	25.1	-36.1	26.0	-38.4	27.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-8.7	-8.4	-3.1	4.9	11.8	16.4	19.2	16.6	11.1	4.9	-1.6	-6.0	(6)	
	DBStd	11.09	7.33	7.66	5.25	4.95	4.82	3.78	3.15	3.40	3.66	4.20	5.41	7.01	(7)	
	HDD10.0	2745	579	516	407	167	36	1	0	1	29	164	348	496	(8)	
	HDD18.3	5034	838	749	666	402	209	82	27	76	218	416	597	754	(9)	
	CDD10.0	858	0	0	0	14	92	193	287	204	62	7	0	0	(10)	
	CDD18.3	108	0	0	0	0	7	24	55	21	1	0	0	0	(11)	
	CDH23.3	742	0	0	0	2	61	184	353	137	4	0	0	0	(12)	
	CDH26.7	163	0	0	0	0	14	40	77	32	0	0	0	0	(13)	
(14) Wind	WSAvg	2.8	2.8	2.9	2.8	2.7	3.0	2.7	2.4	2.5	2.8	3.2	3.0	3.0	(14)	
(15) Precipitation	PrecAvg	661	47	39	35	30	53	74	73	81	65	59	54	50	(15)	
	PrecMax	856	69	77	61	70	124	169	151	178	126	153	109	80	(16)	
	PrecMin	499	17	12	7	13	15	6	8	19	24	21	4	15	(17)	
	PrecStd	99	14	18	13	14	22	38	41	43	28	28	28	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	4.1	10.2	22.3	28.6	30.1	31.2	30.0	23.5	17.8	9.7	7.1	(19)	
		MCWB	2.4	2.3	5.7	13.6	18.8	20.6	22.4	21.9	17.4	12.8	8.2	5.9	(20)	
	2%	DB	2.1	2.7	7.0	18.8	24.6	27.7	28.7	26.6	20.6	13.9	7.6	4.6	(21)	
		MCWB	1.2	1.2	3.6	12.0	17.0	20.3	21.1	19.7	16.1	11.0	6.4	3.7	(22)	
	5%	DB	1.3	1.7	4.8	15.6	21.9	25.5	27.0	24.4	18.5	12.0	6.1	2.9	(23)	
		MCWB	0.5	0.5	2.3	9.9	15.3	19.2	20.3	18.6	14.7	9.9	5.1	1.9	(24)	
	10%	DB	0.5	1.0	3.3	12.9	19.6	23.2	25.2	22.5	16.7	10.6	4.7	1.6	(25)	
		MCWB	-0.2	0.0	1.4	8.3	13.8	17.7	19.3	17.7	13.6	9.0	3.8	0.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	2.7	6.2	14.7	20.7	22.8	23.5	23.0	18.9	13.4	8.6	6.3	(27)	
		MCDB	3.0	3.6	9.7	21.8	26.8	28.2	29.6	28.4	22.4	16.7	9.6	7.2	(28)	
	2%	WB	1.4	1.6	4.1	12.1	17.7	21.1	21.9	20.7	16.7	11.5	6.6	3.8	(29)	
		MCDB	1.8	2.4	6.2	18.0	23.8	26.6	27.3	25.0	19.6	13.5	7.3	4.6	(30)	
	5%	WB	0.7	0.8	2.7	10.4	16.0	19.8	20.9	19.4	15.2	10.2	5.3	2.0	(31)	
		MCDB	1.2	1.6	4.3	14.8	20.6	24.3	25.9	23.5	17.8	11.7	6.0	2.7	(32)	
	10%	WB	-0.2	0.0	1.6	8.6	14.5	18.4	19.9	18.2	13.9	9.1	3.8	0.9	(33)	
		MCDB	0.5	1.0	3.1	12.6	18.7	22.3	24.4	21.8	16.5	10.5	4.6	1.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.5	7.5	8.5	8.6	8.0	8.0	7.8	6.8	4.8	3.7	4.4	(35)	
		MCDBR	4.1	5.1	8.4	13.0	12.6	11.3	11.0	11.3	10.0	7.4	4.7	3.5	(36)	
	5% WB	MCWBR	4.0	4.4	5.7	7.6	6.3	5.5	5.0	5.4	5.7	4.8	4.3	3.4	(37)	
		MCWBR	4.0	4.4	4.9	7.4	6.0	5.4	4.9	5.4	5.4	4.8	4.2	3.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.279	0.322	0.342	0.393	0.374	0.367	0.403	0.404	0.367	0.343	0.307	0.268	(40)	
	taud		2.075	2.063	2.110	2.187	2.383	2.422	2.344	2.319	2.398	2.358	2.185	2.056	(41)	
	Ebn at Noon		533	677	786	809	860	870	825	794	769	672	502	419	(42)	
	Edh at Noon		54	88	112	123	108	106	112	107	85	67	49	40	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.44	1.25	2.69	3.92	5.29	5.51	5.40	4.18	2.48	1.10	0.39	0.23	(44)	
	RadStd		0.07	0.23	0.34	0.42	0.41	0.43	0.63	0.32	0.32	0.18	0.06	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-222	N/A	N/A

Nomenclature: See separate page